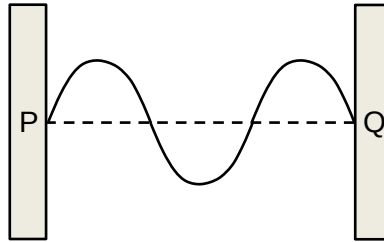
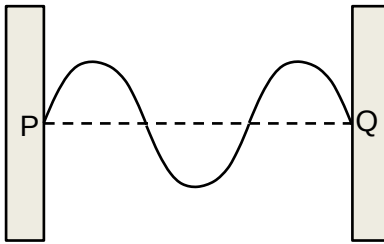


- 18** The diagram shows a stationary wave of frequency 50 Hz formed between two points P and Q at a time $t = 0$.

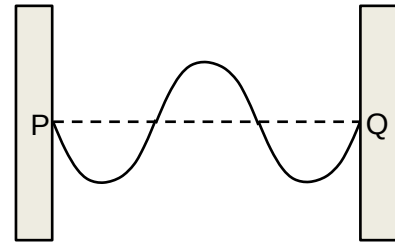


Which of the diagrams correctly shows a possible position of the string at a time $t = 0.010$ s?

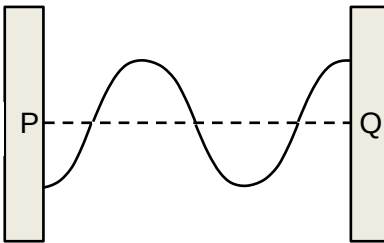
A



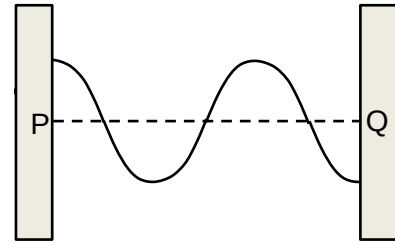
B



C



D



Answer: B

$$f = 50 \text{ Hz} \rightarrow T = 1/50 = 0.020 \text{ s. } t = 0.010 \text{ s} = \frac{1}{2} T$$

- 19** In a spectrometer experiment, light of wavelength 400 nm is incident normally on a diffraction